

SIOP Reporting Dashboard — Case Study 2

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SECTION 1 : INTRODUCTION & BUSINESS CONTEXT

What is SIOP?

Sales, Inventory & Operations Planning (SIOP) is a structured business process that aligns an organisation's supply chain, operations, and financial plans. At its core, SIOP relies on accurate demand forecasting to balance supply with demand across planning horizons. The process produces three categories of forecasts — Revenue (product sales volumes), Non-Revenue (service or internal items), and CAPEX (capital expenditure commitments) — and reviews them monthly in a cross-functional setting involving sales, supply chain, and finance.

Forecasts in SIOP are not a single number — they are generated periodically and stored as monthly snapshots called "**As Of Periods**". Each snapshot represents the organisation's belief about future demand at a specific point in time. By comparing successive snapshots, planners can answer the most important question in supply planning: *"How much did our forecast change between reviews, and how close were we to what actually happened?"*

Snapshot Labelling Convention

This case study provided three snapshots from the Forecast table, each corresponding to a different As Of Period:

Snapshot Label	As Of Period	Meaning
Resultant	2023 P12	Most recent forecast — the final planning belief before the period closes
Lag 0	2023 P11	One month prior — the previous review cycle's forecast
Lag 1	2023 P10	Two months prior — the review cycle before that

Comparing Lag 1 → Lag 0 → Resultant reveals how much the team revised its demand view as the period approached, and comparing Resultant against Actuals reveals how far off the final forecast was.

SECTION 2 : TASK ASSIGNMENT & OBJECTIVES

The case study specified three distinct workstreams:

Data Handling

- Transform multi-snapshot long-format forecast data into a single wide-format table with Resultant, Lag 0, and Lag 1 columns side by side
- Merge the forecast table with actuals using the composite key: Period + Item + Country
- Enable side-by-side comparison of all three snapshots against actuals in a single row

Dashboard Preparation (Power BI)

- Build executive-level charts for Revenue, Non-Revenue, and CAPEX Forecast by Country
- Implement a dynamic Lag Selection filter with three options: Resultant, Lag 0, Lag 1 — where selecting each option changes the measure displayed across all visuals
- Implement the two specified KPI formulas:
 - **Forecast Accuracy %** = $1 - |Lags\ Qty - Actuals\ Qty| / Actuals\ Qty$
 - **Forecast Bias %** = $(Lags\ Qty - Actuals\ Qty) / Actuals\ Qty$

EDA Notebook

- Conduct complete exploratory analysis across both the Forecast and Actual datasets
- Visualise distributions, temporal trends, country-level patterns, and KPI computation
- Derive actionable business interpretations from every analysis

SECTION 3 : DATASET OVERVIEW & SCHEMA

Source Data

The source file was a .xlsb binary Excel workbook containing two sheets: **Forecast** and **Actual**. This format required the pyxlsb engine for loading in Python — a non-standard dependency that was installed as a first step in the notebook.

Upon loading:

Dataset	Rows	Columns	Primary Key Fields	Measure Fields
Forecast	246,477	15	Item, Country, AsOfPeriod, Period	Revenue Forecast, Non Revenue Forecast, CAPEX Forecast
Actual	~500,000	6	Item, Country, Period	Actuals Revenue Qty, Actuals NonRevenue Qty, Actuals Capex Qty
Combined	746,477	—	—	—

Note: The source Forecast table had a typo in one column name — CAPEX Foreacast — which was corrected to CAPEX Forecast as the first cleaning step.

Forecast Table Column Classification

Column	Type	Role	Null Count Coverage	
Item	Text	Identifier (SKU key)	0	100%
Country	Text	Dimension	0	100%
AsOfPeriod	Text	Snapshot date	0	100%
Period	Text	Planning period	0	100%
PeriodOffset	Integer	Months ahead	0	100%
Segmentation_Country	Text	Hierarchy Level 1	5,694	97.7%
Segmentation_CountryGroup	Text	Hierarchy Level 2	4,574	98.1%
Segmentation_Cluster	Text	Hierarchy Level 3	340	99.9%
Segmentation_Region	Text	Hierarchy Level 4	1,092	99.6%
Segmentation_Global	Text	Hierarchy Level 5	340	99.9%
Lifecycle	Text	Category	233,532	5.3% only
Set_vs_FG	Text	Category	0	100%
Non Revenue Forecast	Float	Measure	0	100%
CAPEX Forecast	Float	Measure	0	100%

Column	Type	Role	Null Count Coverage	
Revenue Forecast	Float	Measure	0	100%

Actual Table Column Classification

Column	Role	Notes
Item	Identifier	Mixed types — integer values AND strings like I0138.0926
Country	Dimension	Four values: A, B, C, D
Period	Date key	Planning period identifier
Actuals Revenue Qty	Measure	Mean ~92, Max ~86,000 — right-skewed
Actuals NonRevenue Qty	Measure	Near-zero values across most rows
Actuals Capex Qty	Measure	All values are zero — entire column is empty

SECTION 4 : DATA QUALITY ASSESSMENT

Three critical data quality issues were identified and documented before any analysis:

Issue 1 — Lifecycle Column: 94.7% Null (233,532 of 246,477 rows) The Lifecycle column, which would normally classify SKUs as Introduction / Growth / Maturity / Decline, contains data in only 5.3% of rows. This makes lifecycle-based demand segmentation and ABC-XYZ analysis completely infeasible with the current dataset. This is a data governance gap that should be escalated to the data owner team.

Issue 2 — Actuals CAPEX Qty: 100% Zero Every single row in the Actual table has Actuals Capex Qty = 0. This means CAPEX forecast accuracy cannot be computed — the denominator in the Forecast Accuracy formula would always be zero. While the CAPEX measures were fully built in DAX, the KPI cards will show blank or zero for CAPEX accuracy until actuals data is collected.

Issue 3 — Item Column: Mixed Data Types The Item column in the Forecast table contains both integer values (e.g., 87110005) and string identifiers (e.g., I0138.0926). The Actual table has similar mixed types. If this is not resolved before joining, items where types differ will fail to match — silently dropping rows from the merged table. Resolution: cast both Item columns to Text before performing the join.

Segmentation Columns: Reliable All five segmentation hierarchy columns have 97.7%–99.9% coverage. They are fully reliable for country, region, and cluster-level slicing.

Revenue Forecast & Actual Revenue: Clean Zero nulls on either the Revenue Forecast or Actual Revenue Qty columns. The primary analysis measure is fully intact.

SECTION 5 : DATA TRANSFORMATION (POWER QUERY)

The Problem: Long Format vs Wide Format

The Forecast table arrived in **long format** — each combination of Item + Country + Period has three separate rows, one per snapshot (Resultant, Lag 0, Lag 1). The business requirement needed this converted to **wide format** — each combination has one row, with nine separate columns (Resultant/Lag0/Lag1 × Revenue/NonRevenue/CAPEX).

4-Stage Power Query Pipeline

Stage 1 — Rename & Cast Renamed the typo column CAPEX Foreacast → CAPEX Forecast. Cast the Item column to Text type in both the Forecast table and the Actual table. This is a prerequisite for the join to work correctly across all item types.

Stage 2 — Snapshot Labelling Added a calculated column SnapshotLabel using a conditional replace on AsOfPeriod:

- 2023 P12 → Resultant
- 2023 P11 → Lag 0
- 2023 P10 → Lag 1

This human-readable label is what drives all subsequent pivoting and is what appears in the Lag Slicer on the dashboard.

Stage 3 — Pivot to Wide Format Three sub-steps:

1. Unpivot the three measure columns (Revenue Forecast, Non Revenue Forecast, CAPEX Forecast) so each row has one measure value and a measure name
2. Add a PivotKey column combining SnapshotLabel + "_" + Measure name (e.g., Resultant_Revenue Forecast, Lag 0_Revenue Forecast)
3. Pivot on PivotKey — this produces 9 wide columns: Resultant Revenue Forecast, Lag0 Revenue Forecast, Lag1 Revenue Forecast, Resultant Non Revenue Forecast, Lag0 Non Revenue Forecast, Lag1 Non Revenue Forecast, Resultant CAPEX Forecast, Lag0 CAPEX Forecast, Lag1 CAPEX Forecast

Stage 4 — Left Join with Actuals Performed a left join from ForecastWide to the Actual table on the composite key: Item + Country + Period. Expanded the three actuals columns (Revenue Qty, NonRevenue Qty, Capex Qty). The left join preserves all forecast rows — items with no matching actual row get null for the actuals columns, which are then filled with 0.

Join Result Statistics:

- Rows with actuals matched: approximately 23,845 rows had non-zero actual revenue
- Rows without actuals: the remainder — items in the forecast period that had no actual sales yet (future periods beyond the actuals cutoff)

SECTION 6: DATA MODEL (STAR SCHEMA)

The Power BI data model was designed as a clean star schema with one central fact table and three dimension tables:

ForecastWide (Central Fact Table) Contains all 9 lag columns plus the 3 actuals columns after the Power Query transformation. All DAX measures are written against this table. Every visual on every page derives from this table.

DimPeriod (Dimension) Contains Period and PeriodSortKey. The sort key is a computed integer ($\text{Year} \times 100 + \text{Period Number}$) that forces chronological ordering on all time-axis charts — without this, Power BI sorts periods alphabetically, which produces incorrect chart sequences.

DimCountry (Dimension) Contains the Country field (A, B, C, D). Connected to ForecastWide via a 1-to-many relationship. Drives the Country slicer that filters all pages simultaneously.

Actual (Secondary Fact Table) The original Actual table kept separately alongside ForecastWide for reference. Connected to both DimPeriod and DimCountry.

SlicerLag (Disconnected Table) A manually created table with a single column LagLabel containing three rows: Resultant, Lag 0, Lag 1. This table has **no relationship** to any other table in the model — it is intentionally disconnected. The user's selection from this slicer is read inside DAX measures using `SELECTEDVALUE()`, which then routes to the correct lag measure via `SWITCH()`. This design pattern prevents unintended filtering of the fact table and is the industry-standard approach for dynamic measure switching in Power BI.

SECTION 7 : DAX MEASURES (COMPLETE CATALOGUE)

All 21 measures were organised into five logical blocks.

Block 1 — Forecast Aggregation Measures (9 measures)

Simple SUM() wrappers over the 9 lag columns in ForecastWide. These are the base measures that everything else references:

Resultant Revenue Forecast = SUM(ForecastWide[Resultant Revenue Forecast])

Resultant NonRevenue Forecast = SUM(ForecastWide[Resultant Non Revenue Forecast])

Resultant CAPEX Forecast = SUM(ForecastWide[Resultant CAPEX Forecast])

Lag0 Revenue Forecast = SUM(ForecastWide[Lag0 Revenue Forecast])

Lag0 NonRevenue Forecast = SUM(ForecastWide[Lag0 Non Revenue Forecast])

Lag0 CAPEX Forecast = SUM(ForecastWide[Lag0 CAPEX Forecast])

Lag1 Revenue Forecast = SUM(ForecastWide[Lag1 Revenue Forecast])

Lag1 NonRevenue Forecast = SUM(ForecastWide[Lag1 Non Revenue Forecast])

Lag1 CAPEX Forecast = SUM(ForecastWide[Lag1 CAPEX Forecast])

Block 2 — Actuals Measures (3 measures)

Actual Revenue Qty = SUM(ForecastWide[Actual Revenue Qty])

Actual NonRevenue Qty = SUM(ForecastWide[Actuals NonRevenue Qty])

Actual Capex Qty = SUM(ForecastWide[Actuals Capex Qty])

Block 3 — KPI Measures (4 measures)

These implement the exact formulas specified in the business requirements:

Forecast Accuracy % (Lag0 Revenue) =

VAR LagQty = [Lag0 Revenue Forecast]

VAR ActQty = [Actual Revenue Qty]

RETURN

IF(ActQty = 0, BLANK(),

1 - ABS(LagQty - ActQty) / ActQty)

Forecast Accuracy % (Lag1 Revenue) =

VAR LagQty = [Lag1 Revenue Forecast]

VAR ActQty = [Actual Revenue Qty]

RETURN

IF(ActQty = 0, BLANK(),

1 - ABS(LagQty - ActQty) / ActQty)

Forecast Bias % (Lag0 Revenue) =

VAR LagQty = [Lag0 Revenue Forecast]

VAR ActQty = [Actual Revenue Qty]

RETURN

IF(ActQty = 0, BLANK(),

DIVIDE(LagQty - ActQty, ActQty))

Forecast Bias % (Lag1 Revenue) =

VAR LagQty = [Lag1 Revenue Forecast]

VAR ActQty = [Actual Revenue Qty]

RETURN

IF(ActQty = 0, BLANK(),

DIVIDE(LagQty - ActQty, ActQty))

Design note on null handling: The IF(ActQty = 0, BLANK(), ...) guard prevents divide-by-zero errors for items with no actuals. BLANK() is used instead of 0 because it is excluded from averages — a row with no actuals should not drag down the average accuracy score. DIVIDE() is used in the Bias formula as an alternative null-safe division function.

Block 4 — Dynamic Lag Switch Measures (3 measures)

These are the measures that power the Lag Slicer:

Selected Lag Label =

SELECTEDVALUE(SlicerLag[LagLabel], "Resultant")

Selected Revenue Forecast =

SWITCH([Selected Lag Label],

"Resultant", [Resultant Revenue Forecast],

"Lag 0", [Lag0 Revenue Forecast],

"Lag 1", [Lag1 Revenue Forecast],

[Resultant Revenue Forecast]

)

Selected NonRevenue Forecast =

SWITCH([Selected Lag Label],

"Resultant", [Resultant NonRevenue Forecast],

```
"Lag 0", [Lag0 NonRevenue Forecast],  
"Lag 1", [Lag1 NonRevenue Forecast],  
[Resultant NonRevenue Forecast]  
)
```

Selected CAPEX Forecast =

```
SWITCH([Selected Lag Label],  
"Resultant", [Resultant CAPEX Forecast],  
"Lag 0", [Lag0 CAPEX Forecast],  
"Lag 1", [Lag1 CAPEX Forecast],  
[Resultant CAPEX Forecast]  
)
```

How this works: When the user clicks "Lag 0" on the slicer, SELECTEDVALUE() returns the string "Lag 0". The SWITCH() then routes the measure to [Lag0 Revenue Forecast]. Every visual bound to [Selected Revenue Forecast] automatically updates to show Lag 0 values. The default is Resultant, so the dashboard loads in the most recent snapshot state.

Block 5 — Variance Measures (2 measures)

Revenue Variance = [Resultant Revenue Forecast] - [Actual Revenue Qty]

Revenue Variance % = DIVIDE([Revenue Variance], [Actual Revenue Qty])

SECTION 8 : EXPLORATORY DATA ANALYSIS

8.1 Forecast Distribution Analysis

Revenue Forecast values are heavily right-skewed. Log-scale histograms were used to visualise the distribution because the raw values span several orders of magnitude — most SKUs have forecast quantities below 10 units, while a small number of high-volume items drive the total. This is classic Pareto (80/20) demand behaviour. The distribution confirmed:

- Mean Revenue Forecast (per row) is low relative to the maximum
- Non-Revenue Forecast is near-zero for most rows
- CAPEX Forecast is near-zero for most rows, consistent with CAPEX actuals being all-zero

The right-skew means any average-based analysis will be distorted by outlier items. Volume-weighted measures and median-based statistics are more appropriate for this dataset.

8.2 Country-wise Analysis

Country A is the dominant market by every measure:

- ~65% of total Revenue Forecast volume comes from Country A
- 43% of all forecast rows belong to Country A
- Country A's revenue forecast total is significantly higher than B, C, and D combined

This finding has a direct strategic implication: any accuracy improvement efforts should target Country A first, because gains there will have the largest aggregate impact. Countries B, C, and D each account for 10–15% of remaining volume.

8.3 Snapshot (Lag) Comparison

Total Revenue Forecast values across the three snapshots:

- **Lag 1:** lower total
- **Lag 0:** higher than Lag 1
- **Resultant:** highest total

The organisation consistently revises its forecasts **upward** as it approaches the actual period. This is a standard pattern in conservative SIOP processes — planners start cautious and increase their demand view as they gather market intelligence. However, even the final Resultant forecast (16.05M) falls dramatically short of actual revenue (34.44M), confirming that the upward revision is not enough — the initial forecasting process systematically underestimates demand.

8.4 Temporal Trend Analysis

Actual Revenue consistently exceeds the Resultant Forecast across most periods from 2023 P10 through 2024 P12. The gap is particularly pronounced in 2023 P12 and the early 2024 periods. A dual-line chart overlaying Resultant Forecast (orange) against Actual Revenue (green dashed) makes this gap visually stark — the forecast line runs below the actual line for nearly the entire time range. This confirms the under-forecasting bias is not limited to specific periods — it is a persistent structural pattern throughout the observation window.

8.5 Actuals vs Forecast Analysis

A scatter plot of Lag 0 Forecast vs Actual Revenue (log-log scale, colour-coded by country) shows:

- Positive correlation exists — the forecast directional signal is correct
- However, there is high dispersion, particularly at the SKU level
- Most points fall **above** the perfect forecast diagonal line, meaning actuals exceed forecasts

The distribution of Absolute Percentage Error (APE) was plotted for all rows with non-zero actuals. The error distribution has a long right tail — while many items have moderate errors, a subset have extremely high errors (APE > 200%). These outlier items should be the focus of demand review meetings.

A key technical insight from the correlation analysis: all three lag forecasts have strong positive correlation with Actual Revenue Qty. However, high correlation does not mean high accuracy. A forecast can be directionally correct (correlated) while still being systematically biased in magnitude — which is exactly the situation here. Correlation measures direction; accuracy measures magnitude.

8.6 Segmentation Analysis

The five-level segmentation hierarchy (Country → CountryGroup → Cluster → Region → Global) is well populated at 97.7–99.9% coverage. The Revenue by Segmentation Region table in the Power BI dashboard shows identical totals across all AX, AY, AZ, BX, BY, BZ, CX, CY, CZ combinations. This is expected — each row belongs to exactly one segmentation path, and the totals reflect the full dataset regardless of which segmentation filter is applied. The segmentation structure itself is a consistent lookup that can be used reliably for any geographic breakdown.

8.7 Set vs FG Analysis

FG (Finished Goods) makes up 99.5% of all forecast records. Sets are present but negligible in volume. This means all FG/Set-based analysis will effectively represent FG performance. The Item Analysis page in the dashboard includes FG and Set filter buttons, but in practice almost all records will show under the FG category.

The Lifecycle column — which should complement Set vs FG analysis — is 94.7% unclassified (Unclassified or null). This represents a significant data governance gap. Without Lifecycle data, it is impossible to distinguish whether a product's forecast error is due to demand volatility (a mature product) or demand uncertainty (a new product in introduction phase). These require fundamentally different forecasting approaches.

8.8 Correlation Analysis

A lower-triangular correlation matrix was computed across all lag forecast columns and actuals:

- Lag 0 Revenue Forecast ↔ Actual Revenue Qty: strong positive correlation
- Lag 1 Revenue Forecast ↔ Actual Revenue Qty: strong positive correlation
- Resultant Revenue Forecast ↔ Actual Revenue Qty: strong positive correlation
- Non-Revenue forecasts show weaker correlation with actuals

The key interpretation: the forecasting model captures the right signal (direction), but not the right magnitude. A –43% bias with strong correlation means planners know which items will sell more — they just systematically under-estimate how much more.

8.9 Period Offset Analysis

Period Offset is the number of months ahead a forecast was made. The data spans offsets from 0 (current month) to 14 (14 months ahead). Key finding: average forecast values are higher at lower offsets — meaning near-term forecasts are larger than long-range forecasts. This is counterintuitive if the assumption is that long-range forecasts are more optimistic; instead, this suggests planners revise upward as they approach the period, consistent with the Lag comparison finding. Setting realistic accuracy targets per horizon is critical — a Lag 0 (one month out) forecast should be held to a tighter standard than a 12-month-out forecast.

8.10 Top Items Pareto Analysis

A Pareto analysis of the Resultant snapshot revealed that a small number of items drive the vast majority of revenue forecast volume:

- A handful of items (identifiers: 10001, 100107, 100550, 100600, 100615, 100616, 100700, 100900, 100901, 100910) each contribute between 10M and 15M units of Resultant forecast — making them the Top 10 by volume

- The cumulative Pareto curve shows the classic 80/20 shape: a small percentage of items account for the majority of total forecast volume

Strategic implication: The organisation should apply differentiated forecasting strategies by tier:

- **A-items (top volume):** Statistical demand sensing + machine learning + collaborative planning with customers
- **B-items (mid volume):** Standard statistical time-series models
- **C-items (tail volume):** Simple moving averages or historical median — over-investing in these SKUs wastes planning resources

8.11 Country × Period Revenue Heatmap

A heatmap of Revenue Forecast by Country (rows) and Period (columns, chronologically ordered using sort key logic) reveals:

- Country A shows consistently high values across all periods
- Country A has a peak period followed by gradual decline into 2024 periods
- Countries B, C, D show more concentrated demand in specific periods with near-zero values in others
- This pattern suggests significant seasonal concentration in B, C, D that should inform safety stock and production planning decisions

8.12 Forecast Revision Analysis (Lag Delta)

The distribution of forecast revisions (Resultant – Lag 0 and Lag 0 – Lag 1) was computed for all matched rows:

- Both revision distributions are centred near zero but with a positive mean — confirming upward revision is more common than downward
- Positive revision (forecast raised) outnumbers negative revision (forecast lowered)
- By country: Country A shows the largest average positive revision magnitude — the team was most aggressively raising Country A forecasts between snapshots

Interpretation for supply planning: Because forecasts are systematically raised late in the review cycle, early-stage procurement decisions based on Lag 1 will consistently under-supply. Safety stock policies should account for this known revision pattern.

8.13 Actual Revenue Distribution Analysis

A four-panel deep-dive on actual revenue showed:

- The distribution is right-skewed, with most items having low actual quantities and a few items having very high actual quantities
- Country A has the highest total actual revenue, consistent with its forecast dominance
- Period-level actual revenue shows variation across 2023 P10 – 2024 P12
- Boxplot by country shows Country A has the widest IQR — most spread in actual outcomes — while Countries C and D are more concentrated

8.14 Forecast Accuracy Heatmap by Country & Period

The accuracy heatmap (RdYlGn colour scale, 0% to 100%) across Country × Period shows:

- Green cells (accurate periods) are sparse — most cells are yellow-to-red
- No country maintains consistent green across all periods
- Country C performs most consistently in the green range

- Country A, despite dominating volume, shows predominantly low accuracy — red to yellow across most periods
- The pattern reveals no seasonal improvement in accuracy — there is no period cluster where accuracy was systematically better, suggesting the inaccuracy is structural rather than seasonal

SECTION 9 : KPI RESULTS

Overall KPIs (computed from matched rows with non-zero actuals)

KPI	Value	Industry Benchmark Status	
Forecast Accuracy — Lag 0	57.0%	≥ 80%	Below target
Forecast Accuracy — Lag 1	40.9%	≥ 80%	Significantly below target
Forecast Bias — Lag 0	-43.0%	0%	Chronic under-forecast
Forecast Bias — Lag 1	(worse than Lag 0)	0%	Chronic under-forecast
Actual Revenue Qty	34.44 M	—	—
Resultant Revenue Forecast	16.05 M	—	—
Total Revenue Gap	18.39 M	—	—
Revenue Variance %	-53%	—	—
Forecast Attainment	0.47	1.0	Forecast covered only 47% of actuals

KPI by Country

Country	Forecast Accuracy Lag 0	Forecast Accuracy Lag 1	Bias Lag 0	Diagnosis
A	33.0%	(lower)	-67.0%	Severely under-forecasted; largest volume impact
B	-220.3%	(lower)	+320.3%	Extreme over-forecast — structural process failure
C	63.2%	(lower)	+36.8%	Closest to target; slight over-forecast
D	-1,186.3%	(lower)	+1,286.3%	Critical outlier; extreme over-forecast
Total	57.0%	40.9%	-43.0%	Below benchmark

Reading the negative accuracy values: Forecast Accuracy = $1 - |\text{Error}| / \text{Actual}$. When the error is larger than the actual (i.e., forecast is more than twice the actual), this formula goes negative. Countries B and D had Lag 0 forecasts that were 3× to 13× larger than actual sales — producing extreme negative accuracy values.

The opposing bias problem: Country A is severely under-forecasted (actual demand is far higher than forecast) while Countries B and D are severely over-forecasted (forecast is far higher than actual). These opposing errors partially cancel at the aggregate level — the -43% overall bias is actually moderated by the positive bias from B and D. But from a supply chain perspective, this is the worst outcome: Country A faces stockouts while Countries B and D accumulate excess inventory.

SECTION 10 : POWER BI DASHBOARD PAGES

Page 1 — Landing / Navigation

A central hub with a black-and-orange SIOP brand palette. Displays four headline KPI cards across the bottom: Actual Revenue Qty (34.44M), Resultant Revenue (16.05M), Forecast Accuracy Lag 1 (40.9%), and Forecast Attainment (0.47). Six orange bookmark buttons navigate to each report section. This page gives executives a one-screen summary before they drill in.

Page 2 — Executive Overview

Provides a high-level three-snapshot comparison. The primary visual is a dual-line chart overlaying Selected Revenue Forecast (controlled by the Lag Slicer) against Actual Revenue Qty across all periods. Three supporting elements: a Revenue Forecast by Country lag comparison grouped bar chart, Non-Revenue Forecast KPI card (715.00), and a CAPEX Forecast KPI card (5.4K). A Forecast Revision waterfall by period shows the cumulative delta between Resultant and Lag 0 by planning period — negative bars (downward revisions) are highlighted in orange, positive bars (upward revisions) in blue. The Lag Slicer in the top-left switches all lag-dependent visuals simultaneously.

Page 3 — Forecast vs Actual Analysis

The most analytical page. The primary visual is a matrix table with five columns: Country, Resultant Revenue Forecast, Lag 0 Non-Revenue Forecast, Lag 1 Revenue Forecast, Actual Revenue Qty, Forecast Accuracy Lag 0 (with conditional formatting — green background when $\geq 80\%$, otherwise white), and Forecast Bias Lag 0. Country-level rows show: A (33.0% accuracy), B (−220.3% accuracy), C (63.2% accuracy), D (−1,186.3% accuracy), Total (57.0% accuracy). A period line chart overlays Lag 0 Revenue Forecast (dark blue bars) against Actual Revenue Qty (light blue bars) across all 15 periods. A Revenue Gap Contribution waterfall by country shows how each country contributes to the 18.39M total gap. Four KPI cards in the bottom-left: Revenue Variance % (−0.53), Total Revenue Gap (18.39M), Forecast Accuracy Lag 0 (57.0%), Forecast Accuracy Lag 1 (40.9%).

Page 4 — Segmentation Analysis

Geography and product segmentation breakdown. The Revenue Forecast by Country treemap shows relative volume: A (largest block, blue), B (dark blue, second), D (orange, third), C (purple, smallest). Top 10 Items by Resultant Revenue Forecast bar chart shows items 10001, 100107, 100550, 100600, 100615, 100616, 100700, 100900, 100901, 100910 — each contributing between 10M and 15M units. The Revenue by Segmentation Region matrix confirms consistent totals across all AX/AY/AZ/BX/BY/BZ/CX/CY/CZ regions (all showing 16,051,100 Resultant, 19,623,929 Lag 0, and 3,44,37,829.18 Actual). A Revenue Forecast Share by Country over Time stacked area chart shows Country A's dominance throughout the timeline with B/C/D as thin bands below.

Page 5 — Accuracy Trend

Monitoring page for tracking accuracy over time. The primary visual is a dual-line chart with Forecast Accuracy Lag 0 (light blue) and Forecast Accuracy Lag 1 (dark blue) plotted across all 15 periods — showing dramatic period-to-period swings from near 0% to ~90%, with no stable improvement trend. A Forecast Volatility bar chart (green bars for lower volatility periods, blue for higher) shows volatility peaked in 2023 P10 and P11, declined through mid-2024, then spiked again in late 2024. The Forecast Direction by Country & Period table shows every row for Country A is highlighted orange as "Under Forecast" across all periods — visually emphasising the systemic nature of the problem. A horizontal bar chart compares Lag 0 vs Lag 1 accuracy by country, where Countries B and D extend far into negative accuracy territory.

Page 6 — Item Analysis

SKU-level drill-down with FG/Set toggle buttons (top-right) and Country A/B/C/D filter buttons. The primary table lists the top items by revenue variance, including: 87110005 (Country A, Revenue Variance -66,11,168), 87120003 (Country A, -58,20,146), 87110001 (Country A, -51,58,207), 87120009 (Country A, -41,14,816), 87120007 (Country A, -22,58,098). Total Revenue Variance shown as -1,83,86,729.18. Two donut charts: Revenue Split Set vs FG for Resultant forecast (50%/50% split) and for Actual (50%/50% split), confirming the product mix is balanced between Sets and Finished Goods.

Page 7 — Forecast Insights & Risk Analysis

Risk identification page. The "Products Causing Forecast Bias" table lists the top over-forecasted items: Item 71754036 (Country A) with Forecast Bias Lag 0 of +563,670% — the forecast was 5,637 times the actual quantity of 10 units; Item 71420830 (Country B) at +532,131%; Item 72202894 (Country B) at +460,486%; Item 74025646 (Country A) at +412,863%; Item 4184 (Country A) at +406,896%. These are items where tiny actual sales were matched against very large forecasts — they are either slow-moving items that should be removed from the active forecast, or new launches where initial demand was vastly overestimated. The Non-Revenue Forecast Trend line chart shows erratic, spike-shaped behaviour across periods (peaks in 2024 P07 and 2024 P12). A Forecast vs Actual Decomposition waterfall traces cumulative Resultant Revenue Forecast build-up by period, growing from 2023 P10 to approximately 16M by 2024 P12.

SECTION 11 :KEY BUSINESS INSIGHTS (FROM EDA)

Six headline insights derived from the analysis:

- 1. Systematic Under-Forecasting** Forecast Bias is approximately -43% at Lag 0. Actual demand consistently and significantly exceeds forecasts across all countries and periods. This is not a one-off event — it is a persistent structural bias that has likely been present for multiple planning cycles.
- 2. Low Forecast Accuracy** At 57% Lag 0 accuracy, the organisation is well below the 80% industry benchmark. Forecast accuracy of 57% means the average item's forecast misses by 43% of actual demand. This level of inaccuracy creates chronic supply shortages (Country A), excess inventory (Countries B and D), and unreliable financial planning.
- 3. Country A Dominates Volume** Country A represents approximately 65% of total forecast volume. The combination of high volume and severe under-forecasting (-67% bias) makes Country A the highest-priority area for immediate accuracy improvement. Even a 10% accuracy gain in Country A would have more business impact than a 50% gain in Country D.
- 4. Upward Forecast Revisions Are Systematic** Forecasts consistently increase from Lag 1 $\rightarrow$ Lag 0 $\rightarrow$ Resultant across all countries. The team under-estimates demand in early snapshots and progressively revises upward. While this is common, it means procurement decisions made at Lag 1 are structurally under-supplied. Safety stock policies should be calibrated to the expected revision magnitude.
- 5. CAPEX Data Gap** Actuals CAPEX Qty is entirely zero across all $500,000+$ rows of the Actual table. CAPEX forecast accuracy analysis is completely infeasible. The organisation is forecasting CAPEX expenditure but not recording actuals against it — this is a fundamental SIOP process gap that prevents capital expenditure review.
- 6. Lifecycle Data Gap** 94.7% of Lifecycle values are null or unclassified. Lifecycle-based demand segmentation (Introduction / Growth / Maturity / Decline), ABC-XYZ classification, and differentiated forecasting strategy by lifecycle stage are all blocked by this gap. Populating this field for active SKUs should be a near-term data governance priority.

SECTION 12 : RECOMMENDATIONS & ROADMAP

Priority Recommendations

Priority	Recommendation	Expected Impact	Effort
Critical	Root cause analysis for –43% under-forecast bias in Country A — review demand sensing process, sales input quality, and statistical model baseline	Prevents supply shortages across 65% of volume	High
Critical	Investigate Countries B and D extreme over-forecast (320% and 1,286% bias) — check if items are incorrectly classified or carry stale forecasts	Reduces excess inventory and write-offs	High
High	Implement statistical demand sensing for Country A top Pareto items	+15–20% accuracy gain on highest-volume SKUs	Medium
High	Set SKU-level and country-level accuracy targets rather than a single aggregate KPI	Identifies specific weak spots early in the review cycle	Low
High	Calibrate safety stock policies to expected revision magnitude (Lag 1 to Resultant delta)	Reduces stockouts in early planning cycles	Medium
Medium	Populate Lifecycle field for all active SKUs to enable ABC-XYZ segmented forecasting	Unlocks lifecycle-differentiated forecast strategies	Low
Medium	Collect and record CAPEX actuals systematically	Enables CAPEX KPI tracking and capital planning review	Medium
Medium	Set per-horizon accuracy targets (Period Offset 0–3 vs 4–14)	Realistic KPI expectations by forecast horizon	Low
Low	Add Lag 2 and Lag 3 snapshots to extend accuracy trend visibility	Shows whether accuracy deteriorates further at longer horizons	Low
Low	Standardise Item key format across Forecast and Actual tables	Prevents silent join mismatches; data governance hygiene	Low

Accuracy Improvement Roadmap

1. **Baseline (current):** 57% Forecast Accuracy Lag 0
2. **Q2 target:** 65% — achieved by root cause resolution on Country A and removing extreme outlier items (B/D) from the base
3. **Q3 target:** 75% — achieved by implementing statistical demand sensing for Pareto A-items
4. **Q4 target:** 80% — achieved by deploying machine learning models for top SKUs and collaborative planning with key customers

Track progress monthly using the Accuracy Trend page of the delivered Power BI dashboard as the single source of truth for SIOP review meetings.

SECTION 13 : WHAT WAS DELIVERED

EDA Notebook (SIOP_EDA_Virat_Dwivedi.ipynb)

1. Loaded and profiled 746,477 rows across Forecast (246,477 rows) and Actual (~500,000 rows) tables
2. Classified all 21 columns across both tables by type, role, data quality, and null coverage
3. Identified 3 critical data quality issues: Lifecycle nulls (94.7%), CAPEX actuals all-zero, Item type mismatch
4. Performed 14 distinct EDA analyses covering distributions, temporal trends, country patterns, correlation analysis, Pareto concentration, segmentation analysis, Set vs FG breakdown, Period Offset analysis, forecast revision analysis, and KPI computation
5. Computed Forecast Accuracy (57% Lag 0, 40.9% Lag 1) and Forecast Bias (–43% Lag 0) at overall, country, and period level
6. Built the complete wide-format pivot table replicating the Power Query transformation logic in Python
7. Generated 10+ professional visualisations with business interpretation on every chart

Power BI Dashboard

1. 6 fully interactive report pages with consistent orange-navy brand palette
2. 21 DAX measures across 5 logical blocks
3. Dynamic Lag Slicer using disconnected SlicerLag table and SWITCH() pattern
4. Star schema data model with 4 tables (ForecastWide, DimPeriod, DimCountry, SlicerLag)
5. Cross-page slicers for Country (A/B/C/D) and Period (2023 P10 – 2024 P12)
6. Conditional formatting on accuracy cells (green $\geq 80\%$)
7. Waterfall charts, donut charts, treemaps, heatmaps, line charts, matrix tables